

# Fraction Products

Every graded set, every week — 5 weeks, 25 two-page sheets, with an answer key after each week.

WEEK 1

5 sets

WEEK 2

5 sets

WEEK 3

5 sets

WEEK 4

5 sets

WEEK 5

5 sets

## HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

# Fraction of a Number

Divide by the bottom, multiply by the top.

## Warm-Up 8 ITEMS • RECALL

1  $\frac{1}{2}$  of 10

\_\_\_\_\_

2  $\frac{1}{3}$  of 9

\_\_\_\_\_

3  $\frac{1}{4}$  of 12

\_\_\_\_\_

4  $\frac{1}{5}$  of 20

\_\_\_\_\_

5  $\frac{1}{2}$  of 30

\_\_\_\_\_

6  $\frac{1}{6}$  of 18

\_\_\_\_\_

7  $\frac{1}{3}$  of 12

\_\_\_\_\_

8  $\frac{1}{4}$  of 16

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

9.  $\frac{2}{3}$  of 12

.....

10.  $\frac{3}{4}$  of 20

.....

11.  $\frac{2}{5}$  of 35

.....

12.  $\frac{5}{6}$  of 24

.....

13.  $\frac{3}{8}$  of 40

.....

14.  $\frac{2}{3}$  of 15

.....

15.  $\frac{2}{5}$  of 45

.....

16.  $\frac{5}{6}$  of 42

.....

17.  $\frac{3}{8}$  of 32

.....



# Fraction × Fraction

Multiply across. Type answers like 1/6.

## Warm-Up 8 ITEMS • RECALL

1  $\frac{1}{2} \times \frac{1}{2}$  — type as a/b

\_\_\_\_\_

2  $\frac{1}{2} \times \frac{1}{3}$

\_\_\_\_\_

3  $\frac{1}{3} \times \frac{1}{3}$

\_\_\_\_\_

4  $\frac{1}{2} \times \frac{1}{4}$

\_\_\_\_\_

5  $\frac{1}{5} \times \frac{1}{2}$

\_\_\_\_\_

6  $\frac{1}{4} \times \frac{1}{4}$

\_\_\_\_\_

7  $\frac{1}{4} \times \frac{1}{2}$

\_\_\_\_\_

8  $\frac{1}{3} \times \frac{1}{4}$

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

9.  $\frac{2}{3} \times \frac{1}{2}$

.....

10.  $\frac{3}{4} \times \frac{2}{3}$

.....

11.  $\frac{2}{5} \times \frac{3}{4}$

.....

12.  $\frac{5}{6} \times \frac{3}{5}$

.....

13.  $\frac{3}{8} \times \frac{4}{9}$

.....

14.  $\frac{2}{3} \times \frac{3}{4}$

.....

15.  $\frac{3}{5} \times \frac{2}{3}$

.....

16.  $\frac{4}{5} \times \frac{5}{8}$

.....

17.  $\frac{2}{3} \times \frac{5}{6}$

.....

18.  $\frac{3}{4} \times \frac{4}{9}$

.....



## Working space



## Error journal

# Scaling

Predict bigger or smaller before you calculate.

## Warm-Up 12 ITEMS • RECALL

1  $6 \times 2$  — bigger or smaller than 6  
\_\_\_\_\_

2  $6 \times 1/2$  — bigger or smaller  
\_\_\_\_\_

3  $6 \times 1$  — bigger, smaller or same  
\_\_\_\_\_

4  $6 \times 1/2$   
\_\_\_\_\_

5  $8 \times 1/4$   
\_\_\_\_\_

6  $10 \times 3/5$   
\_\_\_\_\_

7  $8 \times 1$  — bigger, smaller or same  
\_\_\_\_\_

8  $8 \times 2$  — bigger or smaller  
\_\_\_\_\_

9  $8 \times 1/2$  — bigger or smaller  
\_\_\_\_\_

10  $8 \times 1/2$   
\_\_\_\_\_

11  $8 \times 3/4$   
\_\_\_\_\_

12  $8 \times 2$   
\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Short Core today — do all of them, then turn the page over.

13.  $20 \times 3/4$  — bigger or smaller than 20  
.....

14.  $20 \times 5/4$  — bigger or smaller  
.....

15.  $20 \times 5/4$   
.....

16.  $12 \times 7/6$   
.....

17. Why does  $\times 4/4$  change nothing — type the value of  $4/4$   
.....

18.  $20 \times 3/4$   
.....



# Dividing with Fractions

How many of these fit inside that?

## Warm-Up 11 ITEMS • RECALL

1 How many halves in 1

\_\_\_\_\_

2 How many halves in 3

\_\_\_\_\_

3 How many quarters in 1

\_\_\_\_\_

4 How many quarters in 2

\_\_\_\_\_

5 How many thirds in 2

\_\_\_\_\_

6  $4 \div \frac{1}{2}$

\_\_\_\_\_

7  $3 \div \frac{1}{2}$

\_\_\_\_\_

8  $2 \div \frac{1}{4}$

\_\_\_\_\_

9  $5 \div \frac{1}{2}$

\_\_\_\_\_

10  $1 \div \frac{1}{8}$

\_\_\_\_\_

11  $6 \div \frac{1}{3}$

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

12.  $5 \div \frac{1}{4}$

\_\_\_\_\_

13.  $\frac{1}{2} \div 4$  — type as a/b

\_\_\_\_\_

14.  $\frac{1}{3} \div 2$  — type as a/b

\_\_\_\_\_

15.  $\frac{3}{4}$  as a division — type  $3 \div 4$  as a decimal

\_\_\_\_\_

16.  $7 \div \frac{1}{3}$

\_\_\_\_\_

17.  $\frac{1}{3} \div 2$

\_\_\_\_\_

18.  $\frac{1}{4} \div 3$

\_\_\_\_\_

19.  $8 \div \frac{1}{5}$

\_\_\_\_\_

20.  $\frac{1}{2} \div 5$

\_\_\_\_\_



## Working space



## Error journal

# Half a Garden

Enrichment. Fractions of real areas.

NAME \_\_\_\_\_

DATE \_\_\_\_\_

RIGHT \_\_\_\_ OF \_\_\_\_ TRIED

✦ Warm-Up 4 ITEMS • RECALL

- 1 Half of a 12 m<sup>2</sup> bed  
\_\_\_\_\_
- 2 A quarter of 20 m<sup>2</sup>  
\_\_\_\_\_
- 3 1/2 of 20  
\_\_\_\_\_
- 4 1/4 of 20  
\_\_\_\_\_

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. 2/3 of a 18 m<sup>2</sup> bed  
.....
6. A bed 1/2 m by 3/4 m — area as a/b  
.....
7. 2/3 of 24  
.....
8. 3/4 × 2/5 — type as a/b  
.....

 **Challenge** NOT GRADED • REASON IT OUT

9.  $\frac{3}{5}$  of a  $45 \text{ m}^2$  plot

10. A  $\frac{2}{3}$  m by  $\frac{3}{4}$  m bed — area as  $a/b$

11. How many  $\frac{1}{4} \text{ m}^2$  tiles cover  $3 \text{ m}^2$

12. A  $\frac{1}{2}$  by  $\frac{3}{4}$  plot — the area as  $a/b$

13. That plot of a 40 m<sup>2</sup> garden — square metres

14.  $6 \div \frac{1}{4}$

 **Working space** SHOW THE ROOMS, THE GRID, THE NUMBER LINE

**Error journal** ONE SENTENCE PER MISTAKE

Mark the reasoning, not the handwriting.

# Week 1 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

## 1.1 • Fraction of a Number

OUT OF 17

Warm-Up • 1-8 • 5 | 3 | 3 | 4 | 15 | 3 | 4 | 4

Core • 9-17 • 8 | 15 | 14 | 20 | 15 | 10 | 18 | 35 | 12

Challenge • 18-21 • 30 | 175 | 60 | 84

## 1.2 • Fraction × Fraction

OUT OF 18

Warm-Up • 1-8 •  $\frac{1}{4}$  |  $\frac{1}{6}$  |  $\frac{1}{9}$  |  $\frac{1}{8}$  |  $\frac{1}{10}$  |  $\frac{1}{16}$  |  $\frac{1}{8}$  |  $\frac{1}{12}$

Core • 9-18 •  $\frac{1}{3}$  |  $\frac{1}{2}$  |  $\frac{3}{10}$  |  $\frac{1}{2}$  |  $\frac{1}{6}$  |  $\frac{1}{2}$  |  $\frac{2}{5}$  |  $\frac{1}{2}$  |  $\frac{5}{9}$  |  $\frac{1}{3}$

Challenge • 19-22 •  $\frac{1}{4}$  |  $\frac{2}{5}$  |  $\frac{3}{4}$  |  $\frac{1}{6}$

## 1.3 • Scaling

OUT OF 18

Warm-Up • 1-12 • bigger | smaller | same | 3 | 2 | 6 | same | bigger | smaller | 4 | 6 | 16

Core • 13-18 • smaller | bigger | 25 | 14 | 1 | 15

Challenge • 19-22 •  $\frac{4}{3}$  | 30 | smaller | 20

## 1.4 • Dividing with Fractions

OUT OF 20

Warm-Up • 1-11 • 2 | 6 | 4 | 8 | 6 | 8 | 6 | 8 | 10 | 8 | 18

Core • 12-20 • 20 |  $\frac{1}{8}$  |  $\frac{1}{6}$  | 0.75 | 21 |  $\frac{1}{6}$  |  $\frac{1}{12}$  | 40 |  $\frac{1}{10}$

Challenge • 21-24 • 12 |  $\frac{1}{12}$  | 24 |  $\frac{1}{24}$

## Fri • Half a Garden

OUT OF 8

Warm-Up • 1-4 • 6 | 5 | 10 | 5

Core • 5-8 • 12 |  $\frac{3}{8}$  | 16 |  $\frac{3}{10}$

Challenge • 9-14 • 27 |  $\frac{1}{2}$  | 12 |  $\frac{3}{8}$  | 15 | 24

# Fold It Twice

Half of a third is a sixth. Do it on paper before you do it on paper.

## Warm-Up 12 ITEMS • RECALL

1  $\frac{1}{2} \times \frac{1}{2}$

2  $\frac{1}{2} \times \frac{1}{3}$

3  $\frac{1}{3} \times \frac{1}{3}$

4  $\frac{1}{2} \times \frac{1}{4}$

5  $\frac{1}{5} \times \frac{1}{2}$

6  $\frac{1}{4} \times \frac{1}{4}$

7  $\frac{1}{2}$  of  $\frac{1}{2}$  — type as a/b

8  $\frac{1}{2}$  of  $\frac{1}{4}$

9  $\frac{1}{3}$  of  $\frac{1}{2}$

10  $\frac{1}{2}$  of  $\frac{1}{3}$

11  $\frac{1}{4}$  of  $\frac{1}{4}$

12  $\frac{1}{2}$  of  $\frac{1}{5}$

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13.  $\frac{1}{2} \times \frac{1}{5}$

14.  $\frac{1}{3} \times \frac{1}{4}$

15.  $\frac{1}{6} \times \frac{1}{2}$

16.  $\frac{1}{3} \times \frac{1}{5}$

17.  $\frac{1}{4} \times \frac{1}{5}$

18.  $\frac{1}{3}$  of  $\frac{3}{4}$

19.  $\frac{2}{3}$  of  $\frac{1}{2}$

20.  $\frac{3}{4}$  of  $\frac{2}{3}$

21.  $\frac{2}{5}$  of  $\frac{1}{2}$

22.  $\frac{3}{5}$  of  $\frac{5}{6}$

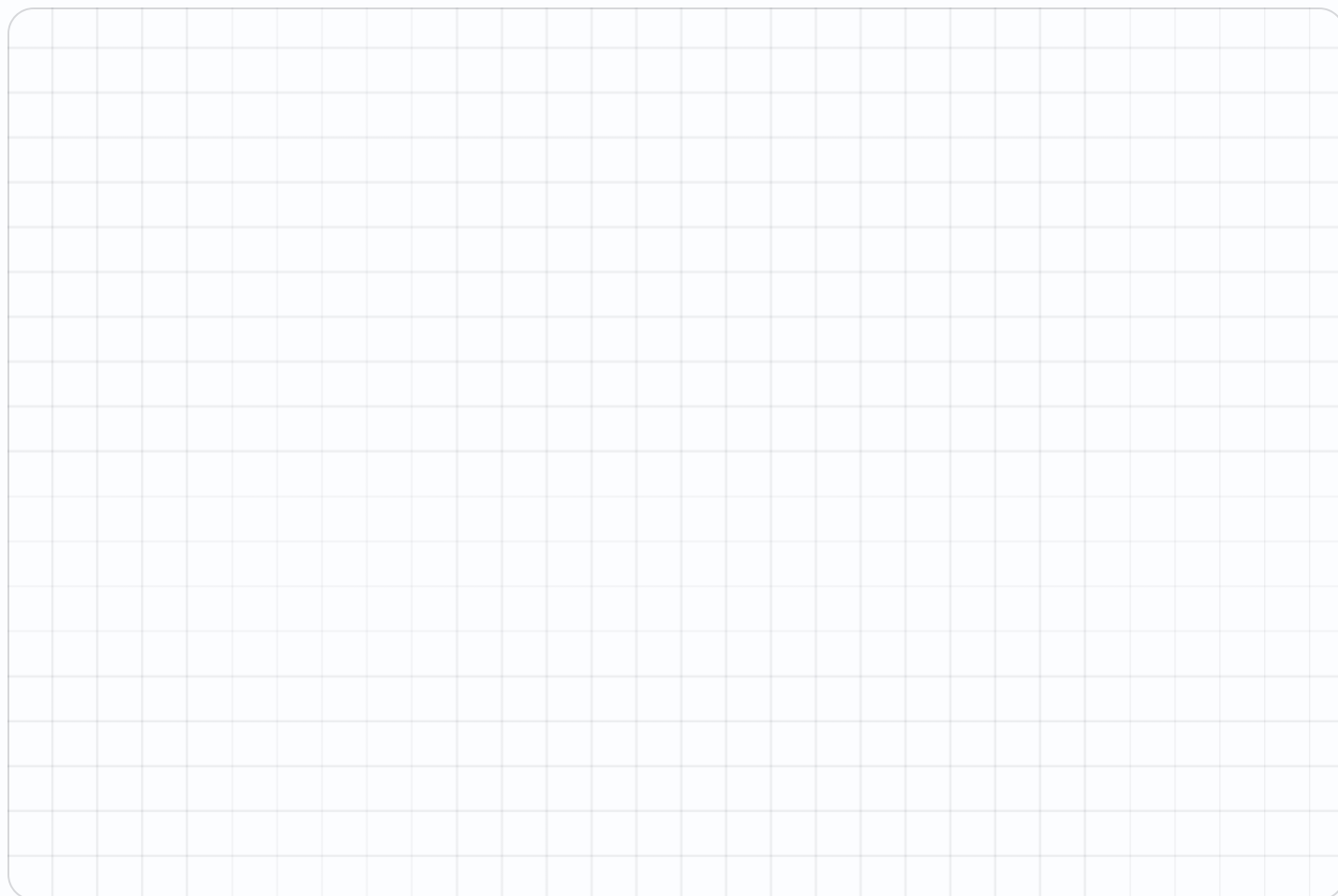
**★ Challenge** NOT GRADED • REASON IT OUT

23.  $\frac{1}{2} \times \frac{1}{3} \times \frac{1}{4}$   
.....

24.  $\frac{1}{8} \times \frac{1}{8}$   
.....

25.  $\frac{2}{3}$  of  $\frac{3}{8}$   
.....

26.  $\frac{5}{6}$  of  $\frac{3}{10}$   
.....

**✎ Working space** SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

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# The Area Model

A rectangle cut both ways. The overlap is the answer.

## Warm-Up 12 ITEMS • RECALL

1 A  $1\frac{1}{2}$  by  $\frac{1}{2}$  square — its area  
\_\_\_\_\_

2 A  $\frac{1}{2}$  by  $\frac{1}{3}$  rectangle  
\_\_\_\_\_

3 A  $\frac{1}{3}$  by  $\frac{1}{4}$  rectangle  
\_\_\_\_\_

4 Pieces when a square is cut in halves both ways  
\_\_\_\_\_

5 Cut in thirds both ways  
\_\_\_\_\_

6 Cut in halves and thirds  
\_\_\_\_\_

7 A 1 by 2 rectangle — area  
\_\_\_\_\_

8 A 2 by 3 rectangle  
\_\_\_\_\_

9 A  $\frac{1}{2}$  by 1 strip — type as a/b  
\_\_\_\_\_

10 A  $\frac{1}{2}$  by  $\frac{1}{2}$  square  
\_\_\_\_\_

11 A  $\frac{1}{3}$  by 1 strip  
\_\_\_\_\_

12 A  $\frac{1}{4}$  by  $\frac{1}{2}$  rectangle  
\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. A  $\frac{2}{3}$  by  $\frac{1}{2}$  rectangle  
.....

14. A  $\frac{3}{4}$  by  $\frac{2}{3}$  rectangle  
.....

15. A  $\frac{2}{5}$  by  $\frac{3}{4}$  rectangle  
.....

16. A  $\frac{2}{3}$  by  $\frac{3}{5}$  rectangle  
.....

17. A  $\frac{5}{6}$  by  $\frac{3}{5}$  rectangle  
.....

18. A  $\frac{2}{3}$  by  $\frac{3}{4}$  rectangle — type as a/b  
.....

19. A  $\frac{3}{5}$  by  $\frac{1}{2}$  rectangle  
.....

20. A  $\frac{4}{5}$  by  $\frac{5}{6}$  rectangle  
.....

21. A  $\frac{3}{4}$  by  $\frac{2}{9}$  rectangle  
.....

22. A  $\frac{5}{8}$  by  $\frac{4}{5}$  rectangle  
.....

**★ Challenge** NOT GRADED • REASON IT OUT23. A  $\frac{3}{8}$  by  $\frac{4}{9}$  rectangle24. A  $\frac{2}{3}$  by  $\frac{3}{4}$  by  $\frac{1}{2}$  box — its volume25. A  $\frac{7}{8}$  by  $\frac{4}{7}$  rectangle26. A  $\frac{5}{6}$  by  $\frac{9}{10}$  rectangle**✎ Working space** SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

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# Multiply Across

Numerators, then denominators. Then simplify.

## Warm-Up 9 ITEMS • RECALL

1  $\frac{1}{2} \times \frac{2}{3}$   
\_\_\_\_\_

2  $\frac{2}{3} \times \frac{1}{2}$   
\_\_\_\_\_

3  $\frac{1}{4} \times \frac{2}{3}$   
\_\_\_\_\_

4  $\frac{3}{4} \times \frac{1}{3}$   
\_\_\_\_\_

5  $\frac{2}{5} \times \frac{1}{2}$   
\_\_\_\_\_

6  $\frac{1}{3} \times \frac{3}{4}$   
\_\_\_\_\_

7  $\frac{1}{2} \times \frac{1}{3}$  — type as a/b  
\_\_\_\_\_

8  $\frac{1}{4} \times \frac{1}{5}$   
\_\_\_\_\_

9  $\frac{1}{2} \times \frac{4}{5}$   
\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

10.  $\frac{3}{4} \times \frac{2}{3}$  .....

11.  $\frac{2}{5} \times \frac{3}{4}$  .....

12.  $\frac{5}{6} \times \frac{3}{5}$  .....

13.  $\frac{3}{8} \times \frac{4}{9}$  .....

14.  $\frac{4}{5} \times \frac{5}{8}$  .....

15.  $\frac{3}{4} \times \frac{5}{6}$  .....

16.  $\frac{2}{7} \times \frac{7}{8}$  .....

17.  $\frac{5}{9} \times \frac{3}{5}$  .....

18.  $\frac{4}{9} \times \frac{3}{8}$  .....

19.  $\frac{7}{10} \times \frac{5}{7}$  .....

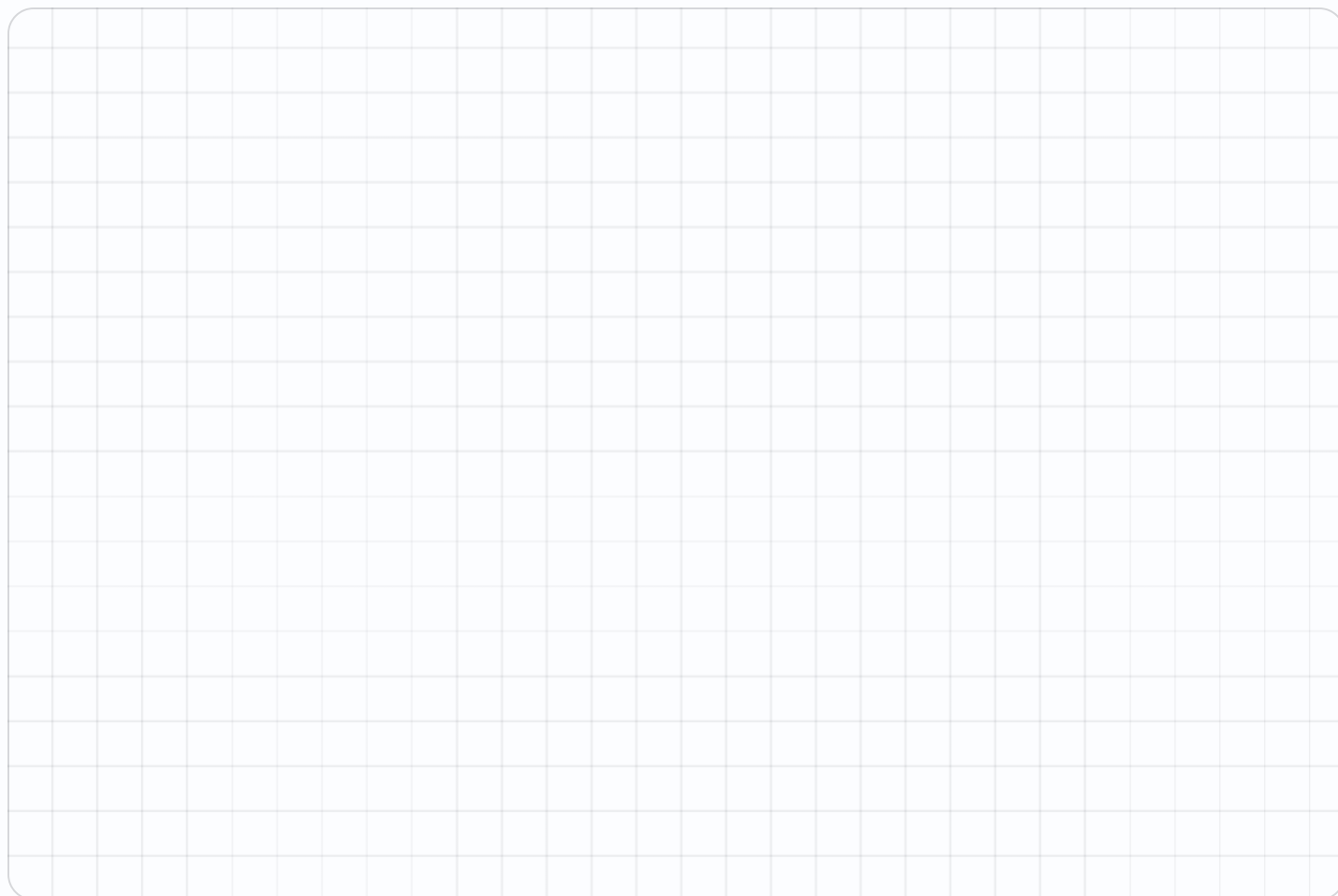
**★ Challenge** NOT GRADED • REASON IT OUT

20.  $\frac{2}{3} \times \frac{3}{4} \times \frac{1}{2}$   
.....

21.  $\frac{6}{7} \times \frac{7}{12}$   
.....

22.  $\frac{9}{16} \times \frac{8}{27}$   
.....

23.  $\frac{11}{12} \times \frac{6}{11}$   
.....

**✎ Working space** SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

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# Simplify as You Go

Cancel before multiplying and the numbers stay small.

## Warm-Up 12 ITEMS • RECALL

1  $2/4$  simplified2  $3/6$ 3  $4/8$ 4  $6/9$ 5  $5/10$ 6  $8/12$ 7  $2/4$  simplified — type as a/b8  $3/6$  simplified9  $4/6$  simplified10  $6/8$  simplified11  $5/10$  simplified12  $9/12$  simplified

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13.  $2/3 \times 3/8$  — simplified14.  $4/9 \times 3/8$ 15.  $5/12 \times 6/10$ 16.  $9/10 \times 5/6$ 17.  $7/8 \times 4/7$ 18.  $2/3 \times 3/5$  simplified19.  $4/5 \times 5/8$  simplified20.  $6/7 \times 7/9$  simplified21.  $3/8 \times 4/9$  simplified22.  $10/11 \times 11/20$  simplified

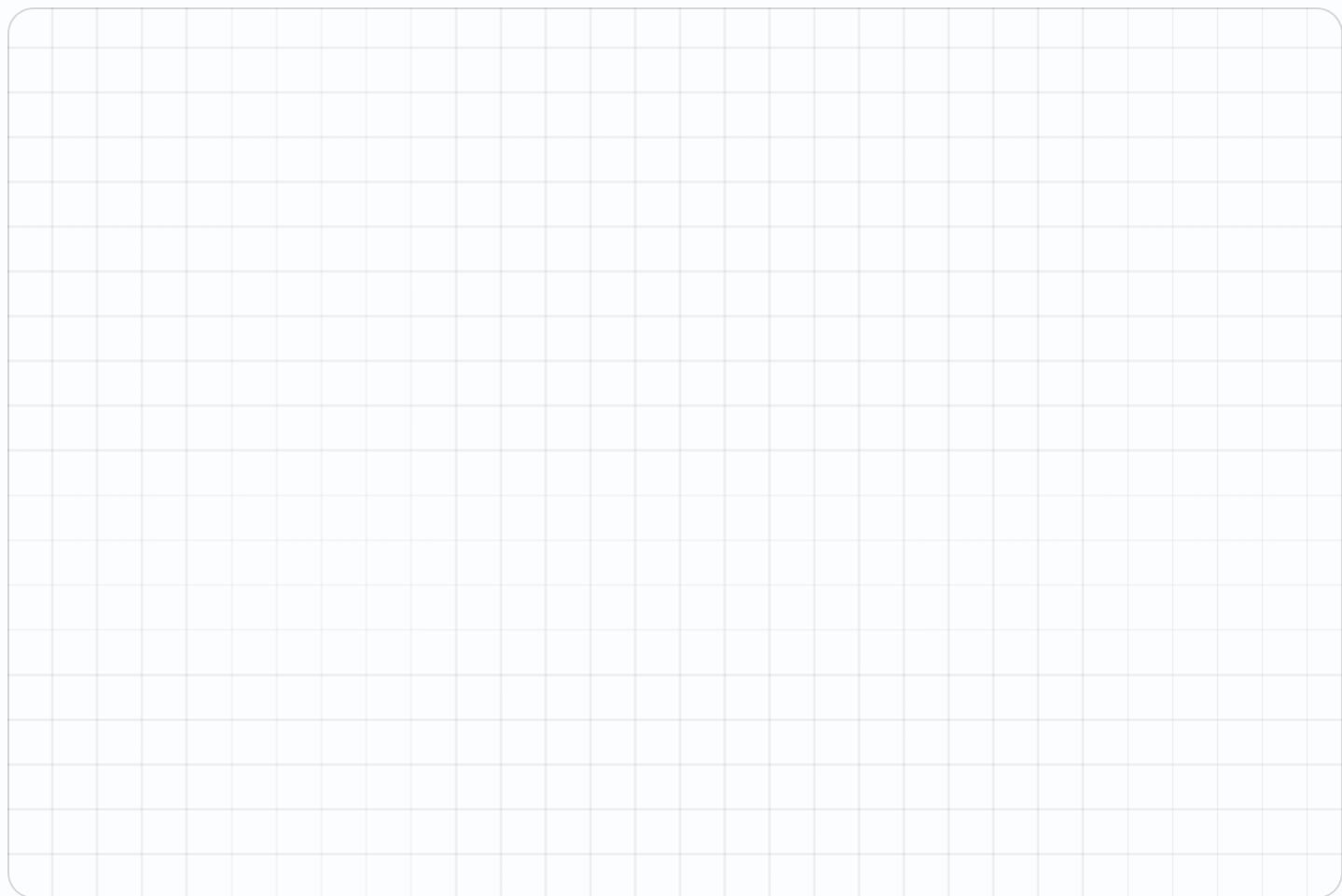
**★ Challenge** NOT GRADED • REASON IT OUT

23.  $12/25 \times 5/6$   
.....

24.  $15/16 \times 8/45$   
.....

25.  $12/25 \times 5/6$  simplified  
.....

26.  $15/16 \times 8/45$  simplified  
.....

**✎ Working space** SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

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# Scaling Showdown

Bigger, smaller or the same — predict before you calculate.

## ✦ Warm-Up 4 ITEMS • RECALL

1  $6 \times 2$  — bigger or smaller than 6  
\_\_\_\_\_

2  $6 \times 1/2$  — bigger or smaller  
\_\_\_\_\_

3  $1/2$  of  $1/2$  — type as a/b  
\_\_\_\_\_

4  $1/2 \times 6$   
\_\_\_\_\_

## ◆ Core GRADED • SHOW YOUR WORKING

**STOP RULE** Short Core today — do all of them, then turn the page over.

5.  $20 \times 3/4$  — bigger or smaller  
.....

6.  $20 \times 5/4$  — bigger or smaller  
.....

7.  $2/3 \times 3/4$  — type as a/b  
.....

8.  $3/5 \times 10$   
.....



# Week 2 Answers

Mark the reasoning, not the handwriting.

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

## 2.1 • Fold It Twice

OUT OF 22

Warm-Up • 1-12 •  $\frac{1}{4}$  |  $\frac{1}{6}$  |  $\frac{1}{9}$  |  $\frac{1}{8}$  |  $\frac{1}{10}$  |  $\frac{1}{16}$  |  $\frac{1}{4}$  |  $\frac{1}{8}$  |  $\frac{1}{6}$  |  $\frac{1}{6}$  |  $\frac{1}{16}$  |  $\frac{1}{10}$

Core • 13-22 •  $\frac{1}{10}$  |  $\frac{1}{12}$  |  $\frac{1}{12}$  |  $\frac{1}{15}$  |  $\frac{1}{20}$  |  $\frac{1}{4}$  |  $\frac{1}{3}$  |  $\frac{1}{2}$  |  $\frac{1}{5}$  |  $\frac{1}{2}$

Challenge • 23-26 •  $\frac{1}{24}$  |  $\frac{1}{64}$  |  $\frac{1}{4}$  |  $\frac{1}{4}$

## 2.2 • The Area Model

OUT OF 22

Warm-Up • 1-12 •  $\frac{1}{4}$  |  $\frac{1}{6}$  |  $\frac{1}{12}$  | 4 | 9 | 6 | 2 | 6 |  $\frac{1}{2}$  |  $\frac{1}{4}$  |  $\frac{1}{3}$  |  $\frac{1}{8}$

Core • 13-22 •  $\frac{1}{3}$  |  $\frac{1}{2}$  |  $\frac{3}{10}$  |  $\frac{2}{5}$  |  $\frac{1}{2}$  |  $\frac{1}{2}$  |  $\frac{3}{10}$  |  $\frac{2}{3}$  |  $\frac{1}{6}$  |  $\frac{1}{2}$

Challenge • 23-26 •  $\frac{1}{6}$  |  $\frac{1}{4}$  |  $\frac{1}{2}$  |  $\frac{3}{4}$

## 2.3 • Multiply Across

OUT OF 19

Warm-Up • 1-9 •  $\frac{1}{3}$  |  $\frac{1}{3}$  |  $\frac{1}{6}$  |  $\frac{1}{4}$  |  $\frac{1}{5}$  |  $\frac{1}{4}$  |  $\frac{1}{6}$  |  $\frac{1}{20}$  |  $\frac{2}{5}$

Core • 10-19 •  $\frac{1}{2}$  |  $\frac{3}{10}$  |  $\frac{1}{2}$  |  $\frac{1}{6}$  |  $\frac{1}{2}$  |  $\frac{5}{8}$  |  $\frac{1}{4}$  |  $\frac{1}{3}$  |  $\frac{1}{6}$  |  $\frac{1}{2}$

Challenge • 20-23 •  $\frac{1}{4}$  |  $\frac{1}{2}$  |  $\frac{1}{6}$  |  $\frac{1}{2}$

## 2.4 • Simplify as You Go

OUT OF 22

Warm-Up • 1-12 •  $\frac{1}{2}$  |  $\frac{1}{2}$  |  $\frac{1}{2}$  |  $\frac{2}{3}$  |  $\frac{1}{2}$  |  $\frac{2}{3}$  |  $\frac{1}{2}$  |  $\frac{1}{2}$  |  $\frac{2}{3}$  |  $\frac{3}{4}$  |  $\frac{1}{2}$  |  $\frac{3}{4}$

Core • 13-22 •  $\frac{1}{4}$  |  $\frac{1}{6}$  |  $\frac{1}{4}$  |  $\frac{3}{4}$  |  $\frac{1}{2}$  |  $\frac{2}{5}$  |  $\frac{1}{2}$  |  $\frac{2}{3}$  |  $\frac{1}{6}$  |  $\frac{1}{2}$

Challenge • 23-26 •  $\frac{2}{5}$  |  $\frac{1}{6}$  |  $\frac{2}{5}$  |  $\frac{1}{6}$

## Fri • Scaling Showdown

OUT OF 8

Warm-Up • 1-4 • bigger | smaller |  $\frac{1}{4}$  | 3

Core • 5-8 • smaller | bigger |  $\frac{1}{2}$  | 6

Challenge • 9-14 • same |  $\frac{4}{3}$  | 30 |  $\frac{3}{4}$  | 14 |  $\frac{1}{2}$

# Bigger or Smaller

Decide without calculating. Then check yourself.

## Warm-Up 9 ITEMS • RECALL

1  $10 \times 2$  — bigger or smaller than 10

\_\_\_\_\_

2  $10 \times 1/2$  — bigger or smaller

\_\_\_\_\_

3  $10 \times 1$  — bigger, smaller or same

\_\_\_\_\_

4  $8 \times 1/4$  — bigger or smaller

\_\_\_\_\_

5  $8 \times 3$  — bigger or smaller

\_\_\_\_\_

6  $8 \times 1/2$

\_\_\_\_\_

7  $10 \times 1/2$

\_\_\_\_\_

8  $10 \times 3/2$

\_\_\_\_\_

9  $10 \times 1/5$

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

10.  $24 \times 2/3$  — bigger or smaller

.....

11.  $24 \times 2/3$

.....

12.  $24 \times 3/2$  — bigger or smaller

.....

13.  $24 \times 3/2$

.....

14.  $12 \times 7/6$

.....

15.  $24 \times 2/3$  — bigger or smaller than 24

.....

16.  $24 \times 4/3$  — bigger or smaller

.....

17.  $24 \times 4/3$

.....

18.  $15 \times 6/5$

.....



## Working space



## Error journal

# Why $\times 1$ Changes Nothing

$\frac{3}{3}$  is one wearing a costume.

## Warm-Up 12 ITEMS • RECALL

1  $\frac{3}{3}$  as a whole number2  $\frac{5}{5}$ 3  $\frac{8}{8}$ 4  $12 \times 1$ 5  $12 \times \frac{3}{3}$ 6  $7 \times \frac{4}{4}$ 7  $5 \times 1$ 8  $5 \times \frac{2}{2}$ 9  $5 \times \frac{3}{3}$ 10  $\frac{2}{2}$  as a whole number11  $\frac{8}{8}$  as a whole number12  $12 \times \frac{4}{4}$ 

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13.  $20 \times \frac{5}{5}$ 14.  $\frac{2}{3} \times \frac{4}{4}$  — type as a/b15.  $\frac{8}{12}$  simplified16.  $15 \times \frac{6}{6}$ 17.  $\frac{3}{4} \times \frac{5}{5}$  — type as a/b18.  $\frac{3}{4} \times \frac{4}{4}$  — type as a/b19. Multiplying by  $\frac{5}{5}$  changes the value — type yes or no20.  $\frac{1}{2} \times \frac{3}{3}$  — type as a/b21. Why rename with  $\frac{3}{3}$  — type same value or new value22.  $\frac{2}{3} = \frac{?}{9}$  — numerator



## Working space



## Error journal

# Compare Products

Which is larger, without working either one out.

## Warm-Up 12 ITEMS • RECALL

1 Larger:  $10 \times 2$  or  $10 \times 3$  — type the multiplier

\_\_\_\_\_

2 Larger:  $10 \times 1/2$  or  $10 \times 1/3$

\_\_\_\_\_

3 Larger:  $8 \times 1$  or  $8 \times 2$

\_\_\_\_\_

4 Larger:  $6 \times 1/4$  or  $6 \times 1/2$

\_\_\_\_\_

5  $10 \times 1/2$

\_\_\_\_\_

6  $10 \times 1/3$  — bigger or smaller than 5

\_\_\_\_\_

7 Bigger — type  $8 \times 2$  or  $8 \times 1/2$

\_\_\_\_\_

8 Bigger — type  $10 \times 1$  or  $10 \times 3/4$

\_\_\_\_\_

9  $8 \times 1/2$

\_\_\_\_\_

10  $8 \times 2$

\_\_\_\_\_

11  $10 \times 3/4$

\_\_\_\_\_

12  $12 \times 1/3$

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Larger:  $20 \times 3/4$  or  $20 \times 4/5$  — type the fraction

.....

14. Larger:  $12 \times 5/6$  or  $12 \times 7/8$

.....

15.  $20 \times 4/5$

.....

16.  $12 \times 7/8$

.....

17. Larger:  $30 \times 2/3$  or  $30 \times 3/5$

.....

18. Without computing, bigger — type  $20 \times 4/5$  or  $20 \times 6/5$

.....

19.  $20 \times 6/5$

.....

20. Bigger — type  $36 \times 5/6$  or  $36 \times 7/6$

.....

21.  $36 \times 5/6$

.....



## Challenge

22. Larger:  $24 \times \frac{5}{8}$  or  $24 \times \frac{7}{12}$

23.  $24 \times \frac{5}{8}$

24. Bigger — type 50 × 9/10 or 50 × 11/10

25. The difference between those two products



## Working space



## Error journal

# Scaling in Context

Recipes, distances and prices, scaled up and down.

## Warm-Up 11 ITEMS • RECALL

1 Half of 20 km

2 Double 15 km

3  $\frac{1}{4}$  of \$404  $\frac{3}{4}$  of 205  $\frac{1}{2}$  of \$186  $\frac{2}{3}$  of 127  $\frac{1}{2}$  of a 20 m plot8  $\frac{1}{4}$  of 40 m9  $\frac{2}{3}$  of 3010  $\frac{1}{5}$  of 5011  $\frac{1}{2}$  of 8 hours

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

12.  $\frac{3}{4}$  of a \$60 price13. A  $\frac{2}{3}$  scaled recipe from 18 cups14.  $\frac{5}{6}$  of a 24 km route15. A price of \$80 at  $\frac{3}{5}$ 16.  $\frac{7}{8}$  of 3217. A recipe scaled by  $\frac{3}{4}$  from 20 cups18. A \$60 price at  $\frac{2}{3}$  — the new price19. A 45-minute lesson at  $\frac{4}{5}$  the length20. A 24 m fence at  $\frac{5}{6}$ 21. A 90-page book,  $\frac{7}{9}$  read — pages read

 **Challenge** NOT GRADED • REASON IT OUT

22. A \$250 item at  $\frac{4}{5}$  of the price

23.  $\frac{3}{8}$  of 96

24. A  $\frac{2}{3}$  scale of a  $\frac{3}{4}$  acre plot — type as a/b

25. That plot in acres if the whole farm is 6 acres

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

## ! Error journal ONE SENTENCE PER MISTAKE

# Measure the Garden

Real beds, real fractions.

NAME \_\_\_\_\_

DATE \_\_\_\_\_

RIGHT \_\_\_\_ OF \_\_\_\_ TRIED

✦ Warm-Up

4 ITEMS • RECALL

- 1 Half of a 12 m<sup>2</sup> bed  
\_\_\_\_\_
- 2 A quarter of 20 m<sup>2</sup>  
\_\_\_\_\_
- 3 A 4 by 5 garden — area  
\_\_\_\_\_
- 4 Half of it  
\_\_\_\_\_

◆ Core

GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. 2/3 of an 18 m<sup>2</sup> bed  
.....
6. A bed 1/2 m by 3/4 m — area  
.....
7. A 1/2 by 3/4 plot — type as a/b  
.....
8. 3/8 of a 40 m<sup>2</sup> garden  
.....

**Challenge** NOT GRADED • REASON IT OUT

9.  $\frac{3}{5}$  of a  $45 \text{ m}^2$  plot

10. A  $\frac{2}{3}$  m by  $\frac{3}{4}$  m bed — area

11. How many  $\frac{1}{4} \text{ m}^2$  tiles cover  $3 \text{ m}^2$

12. A  $\frac{2}{3}$  by  $\frac{5}{6}$  plot — type as a/b

13.  $\frac{5}{9}$  of a  $63 \text{ m}^2$  garden

14. A  $\frac{3}{4}$  by  $\frac{4}{5}$  plot as a fraction of the whole

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

**Error journal** ONE SENTENCE PER MISTAKE

# Week 3 Answers

Mark the reasoning, not the handwriting.

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

## 3.1 • Bigger or Smaller

OUT OF 18

Warm-Up • 1-9 • bigger | smaller | same | smaller | bigger | 4 | 5 | 15 | 2

Core • 10-18 • smaller | 16 | bigger | 36 | 14 | smaller | bigger | 32 | 18

Challenge • 19-22 • smaller | 36 | same | 35

## 3.2 • Why $\times 1$ Changes Nothing

OUT OF 22

Warm-Up • 1-12 • 1 | 1 | 1 | 12 | 12 | 7 | 5 | 5 | 5 | 1 | 1 | 12

Core • 13-22 • 20 |  $8/12$  |  $2/3$  | 15 |  $15/20$  |  $3/4$  | no |  $1/2$  | same value | 6

Challenge • 23-26 • same | 9 |  $7/8$  | 1

## 3.3 • Compare Products

OUT OF 21

Warm-Up • 1-12 • 3 |  $1/2$  | 2 |  $1/2$  | 5 | smaller |  $8 \times 2$  |  $10 \times 1$  | 4 | 16 | 7.5 | 4

Core • 13-21 •  $4/5$  |  $7/8$  | 16 | 10.5 |  $2/3$  |  $20 \times 6/5$  | 24 |  $36 \times 7/6$  | 30

Challenge • 22-25 •  $5/8$  | 15 |  $50 \times 11/10$  | 10

## 3.4 • Scaling in Context

OUT OF 21

Warm-Up • 1-11 • 10 | 30 | 10 | 15 | 9 | 8 | 10 | 10 | 20 | 10 | 4

Core • 12-21 • 45 | 12 | 20 | 48 | 28 | 15 | 40 | 36 | 20 | 70

Challenge • 22-25 • 200 | 36 |  $1/2$  | 3

## Fri • Measure the Garden

OUT OF 8

Warm-Up • 1-4 • 6 | 5 | 20 | 10

Core • 5-8 • 12 |  $3/8$  |  $3/8$  | 15

Challenge • 9-14 • 27 |  $1/2$  | 12 |  $5/9$  | 35 |  $3/5$

# Whole ÷ Unit Fraction

How many of these fit inside that?

## Warm-Up 12 ITEMS • RECALL

1 How many halves in 1

\_\_\_\_\_

2 How many halves in 3

\_\_\_\_\_

3 How many quarters in 1

\_\_\_\_\_

4 How many quarters in 2

\_\_\_\_\_

5 How many thirds in 2

\_\_\_\_\_

6  $4 \div \frac{1}{2}$ 

\_\_\_\_\_

7  $2 \div \frac{1}{2}$ 

\_\_\_\_\_

8  $3 \div \frac{1}{3}$ 

\_\_\_\_\_

9  $4 \div \frac{1}{4}$ 

\_\_\_\_\_

10  $5 \div \frac{1}{5}$ 

\_\_\_\_\_

11  $1 \div \frac{1}{6}$ 

\_\_\_\_\_

12  $2 \div \frac{1}{4}$ 

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13.  $5 \div \frac{1}{4}$ 

.....

14.  $7 \div \frac{1}{3}$ 

.....

15.  $3 \div \frac{1}{8}$ 

.....

16.  $6 \div \frac{1}{5}$ 

.....

17.  $10 \div \frac{1}{2}$ 

.....

18.  $6 \div \frac{1}{3}$ 

.....

19.  $9 \div \frac{1}{2}$ 

.....

20.  $7 \div \frac{1}{4}$ 

.....

21.  $10 \div \frac{1}{5}$ 

.....



## Working space



## Error journal

# Unit Fraction ÷ Whole

Sharing a third between four people.

## Warm-Up 7 ITEMS • RECALL

1  $\frac{1}{2} \div 2$

2  $\frac{1}{2} \div 3$

3  $\frac{1}{3} \div 2$

4  $\frac{1}{4} \div 2$

5  $\frac{1}{5} \div 2$

6  $\frac{1}{3} \div 3$

7  $\frac{1}{2} \div 2$  — type as a/b

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

8.  $\frac{1}{2} \div 4$

9.  $\frac{1}{3} \div 4$

10.  $\frac{1}{4} \div 3$

11.  $\frac{1}{5} \div 4$

12.  $\frac{1}{6} \div 2$

13.  $\frac{1}{2} \div 5$  — type as a/b

14.  $\frac{1}{4} \div 4$

15.  $\frac{1}{6} \div 3$

16.  $\frac{1}{3} \div 5$

17.  $\frac{1}{8} \div 2$



## Working space



## Error journal

# Fraction as Division

$\frac{3}{4}$  is literally  $3 \div 4$ .

## Warm-Up 12 ITEMS • RECALL

1  $1 \div 2$  as a decimal

\_\_\_\_\_

2  $1 \div 4$  as a decimal

\_\_\_\_\_

3  $3 \div 4$  as a decimal

\_\_\_\_\_

4  $1 \div 5$  as a decimal

\_\_\_\_\_

5  $2 \div 5$  as a decimal

\_\_\_\_\_

6  $1 \div 10$  as a decimal

\_\_\_\_\_

7  $6 \div 2$  as a fraction — type as a/b

\_\_\_\_\_

8  $3 \div 4$  — type as a/b

\_\_\_\_\_

9  $1 \div 2$ 

\_\_\_\_\_

10  $5 \div 8$ 

\_\_\_\_\_

11  $2 \div 3$ 

\_\_\_\_\_

12  $7 \div 10$ 

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13.  $3 \div 8$  as a decimal

.....

14.  $5 \div 8$  as a decimal

.....

15.  $7 \div 10$  as a decimal

.....

16. 3 pizzas between 4 — each, as a/b

.....

17. 5 cakes between 2 — each, as a/b

.....

18. 3 pizzas shared by 4 — each share as a/b

.....

19. 5 cakes shared by 8

.....

20. 7 metres shared by 2

.....

21. 9 shared by 4 — type as a/b

.....

22. 11 shared by 3

.....



# Ribbon, Pizza and Time

Real contexts for dividing with fractions.

## Warm-Up 12 ITEMS • RECALL

1 A 2 m ribbon into  $\frac{1}{2}$  m pieces

\_\_\_\_\_

2 A 3 m ribbon into  $\frac{1}{2}$  m pieces

\_\_\_\_\_

3 Half a pizza between 2

\_\_\_\_\_

4 A whole pizza in quarters — slices

\_\_\_\_\_

5 An hour in half hours

\_\_\_\_\_

6 An hour in quarter hours

\_\_\_\_\_

7 A 6 m ribbon into  $\frac{1}{2}$  m pieces

\_\_\_\_\_

8 A 4 m ribbon into  $\frac{1}{4}$  m pieces

\_\_\_\_\_

9 A pizza in  $\frac{1}{8}$  slices — slices

\_\_\_\_\_

10 2 pizzas in  $\frac{1}{8}$  slices

\_\_\_\_\_

11 3 hours in  $\frac{1}{2}$  hours

\_\_\_\_\_

12 A 5 m ribbon into 1 m pieces

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. A 4.5 m rope into  $\frac{1}{2}$  m pieces

.....

14.  $\frac{1}{3}$  of a cake between 2

.....

15. 2 hours in  $\frac{1}{4}$  hours

.....

16. A 5 m ribbon into  $\frac{1}{4}$  m bows

.....

17.  $\frac{1}{2}$  of a pie between 3

.....

18. A 7 m ribbon into  $\frac{1}{4}$  m pieces

.....

19. A  $\frac{1}{2}$  pizza shared by 4 — type as a/b

.....

20. 5 hours in  $\frac{1}{4}$  hours

.....

21. A  $\frac{2}{3}$  m ribbon halved — type as a/b

.....

22. A 9 m ribbon into  $\frac{1}{3}$  m pieces

.....



## Working space



## Error journal

# Mid-Unit Quiz

Eight items across Weeks 1–4. 85% to keep flying.

## Warm-Up 4 ITEMS • RECALL

1  $\frac{1}{2} \times \frac{1}{3}$   
\_\_\_\_\_

2  $\frac{2}{3}$  of 9  
\_\_\_\_\_

3  $\frac{1}{2}$  of 12  
\_\_\_\_\_

4  $4 \div \frac{1}{2}$   
\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** This one is a test. Do **every** problem — no stopping early.

5.  $\frac{3}{4} \times \frac{2}{3}$  .....  
\_\_\_\_\_

6.  $24 \times \frac{2}{3}$  .....  
\_\_\_\_\_

7.  $5 \div \frac{1}{4}$  .....  
\_\_\_\_\_

8.  $\frac{1}{3} \div 4$  .....  
\_\_\_\_\_

9.  $20 \times \frac{3}{4}$  — bigger or smaller than 20 .....  
\_\_\_\_\_

10.  $\frac{2}{3} \times \frac{3}{4}$  — type as a/b .....  
\_\_\_\_\_

11.  $\frac{1}{2} \div 4$  .....  
\_\_\_\_\_

12.  $\frac{3}{4}$  of 20 .....  
\_\_\_\_\_

13.  $6 \div \frac{1}{3}$  .....  
\_\_\_\_\_

14.  $3 \div 4$  — type as a/b .....  
\_\_\_\_\_

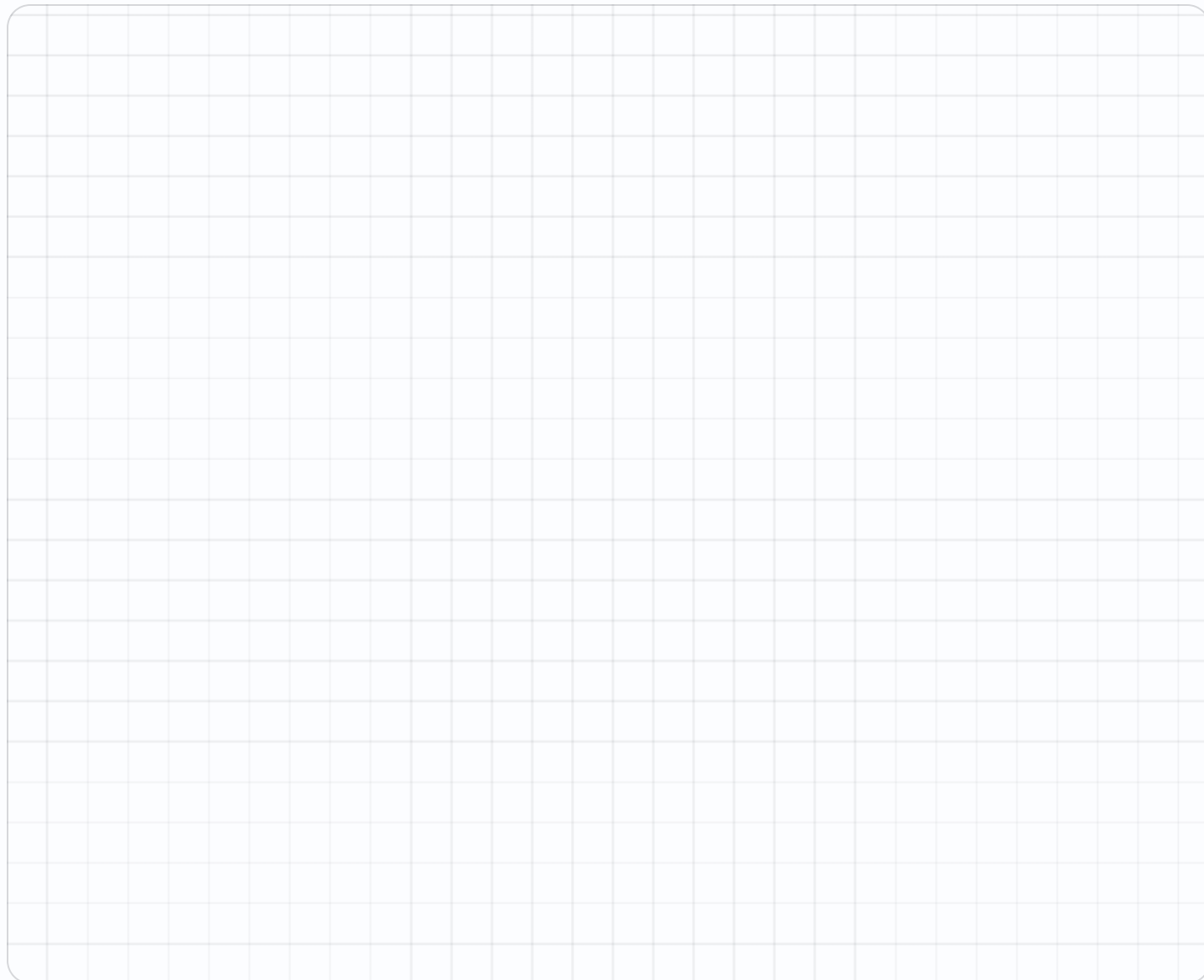
**★ Challenge** NOT GRADED • REASON IT OUT

15.  $\frac{2}{3} \times \frac{3}{4} \times \frac{1}{2}$

.....

16.  $\frac{5}{6} \times \frac{9}{10}$

.....

**✎ Working space** SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

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# Week 4 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

## 4.1 • Whole ÷ Unit Fraction

OUT OF 21

Warm-Up • 1-12 • 2 | 6 | 4 | 8 | 6 | 8 | 4 | 9 | 16 | 25 | 6 | 8

Core • 13-21 • 20 | 21 | 24 | 30 | 20 | 18 | 18 | 28 | 50

Challenge • 22-25 • 12 | 72 | 40 | 36

## 4.2 • Unit Fraction ÷ Whole

OUT OF 17

Warm-Up • 1-7 •  $\frac{1}{4}$  |  $\frac{1}{6}$  |  $\frac{1}{6}$  |  $\frac{1}{8}$  |  $\frac{1}{10}$  |  $\frac{1}{9}$  |  $\frac{1}{4}$

Core • 8-17 •  $\frac{1}{8}$  |  $\frac{1}{12}$  |  $\frac{1}{12}$  |  $\frac{1}{20}$  |  $\frac{1}{12}$  |  $\frac{1}{10}$  |  $\frac{1}{16}$  |  $\frac{1}{18}$  |  $\frac{1}{15}$  |  $\frac{1}{16}$

Challenge • 18-21 •  $\frac{1}{12}$  |  $\frac{1}{32}$  |  $\frac{1}{30}$  |  $\frac{1}{12}$

## 4.3 • Fraction as Division

OUT OF 22

Warm-Up • 1-12 • 0.5 | 0.25 | 0.75 | 0.2 | 0.4 | 0.1 | 3 |  $\frac{3}{4}$  |  $\frac{1}{2}$  |  $\frac{5}{8}$  |  $\frac{2}{3}$  |  $\frac{7}{10}$

Core • 13-22 • 0.375 | 0.625 | 0.7 |  $\frac{3}{4}$  |  $\frac{5}{2}$  |  $\frac{3}{4}$  |  $\frac{5}{8}$  |  $\frac{7}{2}$  |  $\frac{9}{4}$  |  $\frac{11}{3}$

Challenge • 23-26 • 0.875 | 2.25 | 3 | 2.6

## 4.4 • Ribbon, Pizza and Time

OUT OF 22

Warm-Up • 1-12 • 4 | 6 |  $\frac{1}{4}$  | 4 | 2 | 4 | 12 | 16 | 8 | 16 | 6 | 5

Core • 13-22 • 9 |  $\frac{1}{6}$  | 8 | 20 |  $\frac{1}{6}$  | 28 |  $\frac{1}{8}$  | 20 |  $\frac{1}{3}$  | 27

Challenge • 23-26 • 24 | 18 |  $\frac{1}{15}$  | 8

## Fri • Mid-Unit Quiz

OUT OF 14

Warm-Up • 1-4 •  $\frac{1}{6}$  | 6 | 6 | 8

Core • 5-14 •  $\frac{1}{2}$  | 16 | 20 |  $\frac{1}{12}$  | smaller |  $\frac{1}{2}$  |  $\frac{1}{8}$  | 15 | 18 |  $\frac{3}{4}$

Challenge • 15-16 •  $\frac{1}{4}$  |  $\frac{3}{4}$

# Finish the Garden Plan

Every bed with its fraction and its area.

## Warm-Up 12 ITEMS • RECALL

1  $\frac{1}{2}$  of 20 m<sup>2</sup>

\_\_\_\_\_

2  $\frac{1}{4}$  of 24 m<sup>2</sup>

\_\_\_\_\_

3  $\frac{1}{3}$  of 27 m<sup>2</sup>

\_\_\_\_\_

4  $\frac{2}{3}$  of 12 m<sup>2</sup>

\_\_\_\_\_

5  $\frac{3}{4}$  of 16 m<sup>2</sup>

\_\_\_\_\_

6  $\frac{1}{5}$  of 30 m<sup>2</sup>

\_\_\_\_\_

7 A  $\frac{1}{2}$  by  $\frac{1}{2}$  plot — type as a/b

\_\_\_\_\_

8  $\frac{1}{4}$  of a 40 m<sup>2</sup> garden

\_\_\_\_\_

9  $\frac{1}{2}$  of 30 m<sup>2</sup>

\_\_\_\_\_

10 A 2 by 3 bed — area

\_\_\_\_\_

11  $\frac{1}{3}$  of 24 m<sup>2</sup>

\_\_\_\_\_

12  $\frac{3}{4}$  of 12 m<sup>2</sup>

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13.  $\frac{2}{5}$  of 45 m<sup>2</sup>

.....

14.  $\frac{5}{6}$  of 36 m<sup>2</sup>

.....

15. A  $\frac{3}{4}$  m by  $\frac{2}{3}$  m bed — area

.....

16.  $\frac{3}{8}$  of 64 m<sup>2</sup>

.....

17.  $\frac{7}{10}$  of 50 m<sup>2</sup>

.....

18. A  $\frac{2}{3}$  by  $\frac{3}{4}$  plot — type as a/b

.....

19.  $\frac{1}{2}$  of a 48 m<sup>2</sup> garden

.....

20. A  $\frac{3}{5}$  by  $\frac{5}{6}$  plot as a/b

.....

21. A garden of 60 m<sup>2</sup>,  $\frac{2}{5}$  in vegetables — m<sup>2</sup>

.....

22. The remaining share of that garden as a/b

.....

**★ Challenge** NOT GRADED • REASON IT OUT

23. A  $\frac{2}{3}$  m by  $\frac{5}{6}$  m bed — area

.....

24. How many  $\frac{1}{8}$  m<sup>2</sup> tiles cover 2 m<sup>2</sup>

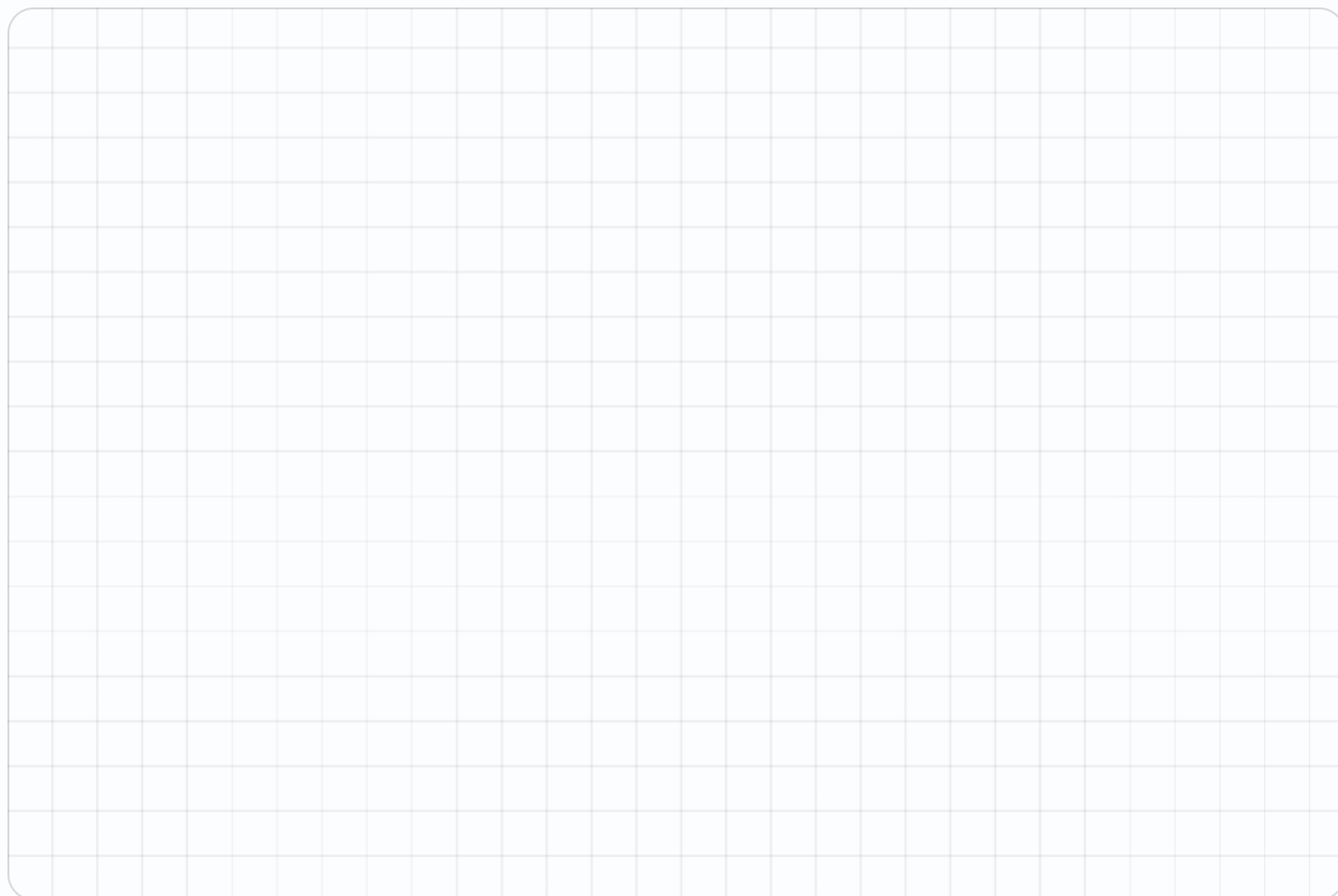
.....

25. A  $\frac{5}{8}$  by  $\frac{4}{5}$  plot as a/b

.....

26. Three beds each  $\frac{2}{9}$  of a 54 m<sup>2</sup> garden — total m<sup>2</sup>

.....

**✎ Working space** SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

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# Explain a Shrink

Name one product smaller than its starting number and say why.

## Warm-Up 11 ITEMS • RECALL

1  $12 \times \frac{1}{2}$

2  $12 \times \frac{1}{3}$

3  $12 \times \frac{1}{4}$

4  $12 \times 2$

5  $12 \times 1$

6  $12 \times \frac{3}{4}$

7  $20 \times \frac{1}{2}$

8  $20 \times \frac{1}{4}$

9 Multiplying by  $\frac{1}{2}$  makes it — type half or double

10  $30 \times \frac{1}{5}$

11  $16 \times \frac{1}{8}$

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

12. Which shrinks 12:  $\times \frac{3}{4}$  or  $\times \frac{5}{4}$  — type the fraction

13.  $30 \times \frac{2}{5}$

14.  $30 \times \frac{5}{2}$

15. Which grows 30 — type  $\frac{2}{5}$  or  $\frac{5}{2}$ 

16.  $40 \times \frac{7}{8}$

17.  $40 \times \frac{3}{4}$  — bigger or smaller than 40

18.  $40 \times \frac{3}{4}$

19. Why is it smaller — type factor under 1 or over 1

20.  $40 \times \frac{5}{4}$

21.  $18 \times \frac{5}{6}$



## Working space



## Error journal

# Multiply and Divide Together

Mixed set. Read before you reach for a method.

## Warm-Up 10 ITEMS • RECALL

1  $\frac{1}{2} \times \frac{1}{2}$

2  $2 \div \frac{1}{2}$

3  $\frac{1}{2}$  of 8

4  $\frac{1}{3} \div 2$

5  $\frac{1}{4} \times \frac{1}{2}$

6  $3 \div \frac{1}{3}$

7  $\frac{1}{2}$  of 16

8  $4 \div \frac{1}{2}$

9  $\frac{1}{2} \times \frac{1}{4}$  — type as a/b

10  $\frac{2}{3}$  of 9

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

11.  $\frac{3}{4} \times \frac{2}{3}$

12.  $5 \div \frac{1}{4}$

13.  $\frac{2}{5}$  of 35

14.  $\frac{1}{4} \div 3$

15.  $24 \times \frac{5}{6}$

16.  $\frac{3}{4} \times \frac{2}{3}$  — type as a/b

17.  $8 \div \frac{1}{4}$

18.  $\frac{5}{6}$  of 42

19.  $20 \times \frac{3}{4}$

**★ Challenge** NOT GRADED • REASON IT OUT

20.  $\frac{2}{3} \times \frac{3}{4} \times \frac{1}{2}$

21. A 3 m ribbon into  $\frac{1}{4}$  m pieces

22.  $\frac{5}{6} \times \frac{9}{10}$

23. How many  $\frac{3}{4}$  m pieces from a 6 m ribbon**✎ Working space** SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

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# Error Journal Sweep

Re-read every entry from the mission. Fix only what repeats.

## Warm-Up 11 ITEMS • RECALL

1  $\frac{1}{2} \times \frac{1}{4}$

2  $\frac{1}{3}$  of 12

3  $4 \div \frac{1}{2}$

4  $\frac{1}{2} \div 2$

5  $\frac{3}{4}$  of 20

6  $\frac{1}{5} \times \frac{1}{2}$

7  $\frac{1}{2}$  of 10

8  $\frac{1}{2} \times \frac{1}{2}$  — type as a/b

9  $4 \div \frac{1}{4}$

10  $\frac{3}{4}$  of 8

11  $2 \div \frac{1}{2}$

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

12.  $\frac{2}{5} \times \frac{3}{4}$

13.  $7 \div \frac{1}{3}$

14.  $\frac{5}{6}$  of 24

15.  $\frac{1}{5} \div 4$

16.  $24 \times \frac{3}{2}$

17.  $\frac{2}{3} \times \frac{3}{5}$  — type as a/b

18.  $7 \div \frac{1}{4}$

19.  $\frac{1}{6} \div 3$

20.  $\frac{3}{8}$  of 32

21.  $24 \times \frac{2}{3}$



# Mission 06 Test

Twelve items plus the Big Question, answered out loud.

## Warm-Up 4 ITEMS • RECALL

1  $\frac{1}{2} \times \frac{1}{3}$   
\_\_\_\_\_

2  $\frac{1}{4}$  of 20  
\_\_\_\_\_

3  $\frac{1}{2}$  of 14  
\_\_\_\_\_

4  $3 \div \frac{1}{2}$   
\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** This one is a test. Do **every** problem — no stopping early.

5.  $\frac{2}{3}$  of 12  
.....

6.  $\frac{3}{4} \times \frac{2}{3}$   
.....

7.  $\frac{2}{5} \times \frac{3}{4}$   
.....

8.  $24 \times \frac{2}{3}$   
.....

9.  $20 \times \frac{5}{4}$   
.....

10.  $5 \div \frac{1}{4}$   
.....

11.  $\frac{1}{3} \div 4$   
.....

12.  $3 \div 4$  as a decimal  
.....

13.  $\frac{1}{2} \times \frac{2}{3}$  — type as a/b  
.....

14.  $\frac{2}{5}$  of 45  
.....

15.  $9 \div \frac{1}{3}$   
.....

16.  $\frac{1}{4} \div 5$   
.....

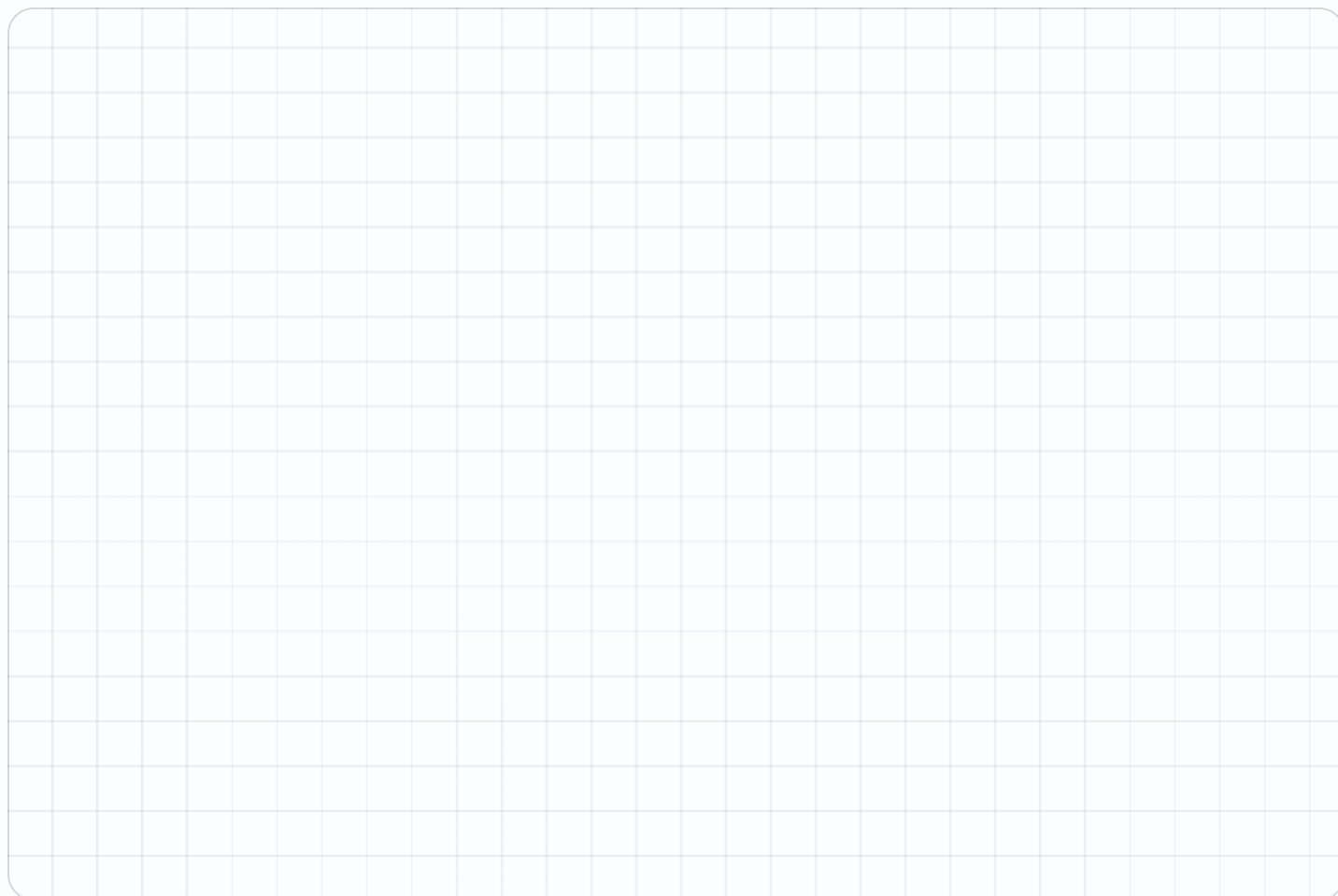
17.  $\frac{3}{4} \times \frac{8}{9}$   
.....

**★ Challenge** NOT GRADED • REASON IT OUT

21.  $\frac{2}{3} \times \frac{3}{4} \times \frac{1}{2}$   
.....

22. A  $\frac{2}{3}$  m by  $\frac{3}{4}$  m bed — area  
.....

23.  $\frac{9}{16} \times \frac{8}{27}$   
.....

24. How many  $\frac{3}{8}$  m pieces from a 3 m ribbon  
.....**✎ Working space** SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

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# Week 5 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

## 5.1 • Finish the Garden Plan

OUT OF 22

Warm-Up • 1-12 • 10 | 6 | 9 | 8 | 12 | 6 |  $\frac{1}{4}$  | 10 | 15 | 6 | 8 | 9

Core • 13-22 • 18 | 30 |  $\frac{1}{2}$  | 24 | 35 |  $\frac{1}{2}$  | 24 |  $\frac{1}{2}$  | 24 |  $\frac{3}{5}$

Challenge • 23-26 •  $\frac{5}{9}$  | 16 |  $\frac{1}{2}$  | 36

## 5.2 • Explain a Shrink

OUT OF 21

Warm-Up • 1-11 • 6 | 4 | 3 | 24 | 12 | 9 | 10 | 5 | half | 6 | 2

Core • 12-21 •  $\frac{3}{4}$  | 12 | 75 |  $\frac{5}{2}$  | 35 | smaller | 30 | factor under 1 | 50 | 15

Challenge • 22-25 • smaller | 50 | 1 | 63

## 5.3 • Multiply and Divide Together

OUT OF 19

Warm-Up • 1-10 •  $\frac{1}{4}$  | 4 | 4 |  $\frac{1}{6}$  |  $\frac{1}{8}$  | 9 | 8 | 8 |  $\frac{1}{8}$  | 6

Core • 11-19 •  $\frac{1}{2}$  | 20 | 14 |  $\frac{1}{12}$  | 20 |  $\frac{1}{2}$  | 32 | 35 | 15

Challenge • 20-23 •  $\frac{1}{4}$  | 12 |  $\frac{3}{4}$  | 8

## Thu • Error Journal Sweep

OUT OF 21

Warm-Up • 1-11 •  $\frac{1}{8}$  | 4 | 8 |  $\frac{1}{4}$  | 15 |  $\frac{1}{10}$  | 5 |  $\frac{1}{4}$  | 16 | 6 | 4

Core • 12-21 •  $\frac{3}{10}$  | 21 | 20 |  $\frac{1}{20}$  | 36 |  $\frac{2}{5}$  | 28 |  $\frac{1}{18}$  | 12 | 16

Challenge • 22-25 •  $\frac{1}{6}$  | 24 |  $\frac{1}{6}$  |  $\frac{1}{15}$

## Fri • Mission 06 Test

OUT OF 20

Warm-Up • 1-4 •  $\frac{1}{6}$  | 5 | 7 | 6

Core • 5-20 • 8 |  $\frac{1}{2}$  |  $\frac{3}{10}$  | 16 | 25 | 20 |  $\frac{1}{12}$  | 0.75 |  $\frac{1}{3}$  | 18 | 27 |  $\frac{1}{20}$   
|  $\frac{2}{3}$  | 24 |  $\frac{5}{8}$  |  $\frac{1}{2}$

Challenge • 21-24 •  $\frac{1}{4}$  |  $\frac{1}{2}$  |  $\frac{1}{6}$  | 8